

Power Series Solutions to Nonlinear Ordinary Differential Equations

MATH 689 Final

Rochester Institute of Technology
Ducheng Tan

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A Mathematical Model for chronic myelogenous leukemia (CML) and T cell interaction [1]

The model we will be solving is a adjusted version of the original system of nonlinear ODEs in Moore and Li's paper from Anne Talkington, Claudia Dantoin, and Rick Durrett. [2]

$$\begin{aligned}\frac{dT}{dt} &= aT(1 - bT) - \gamma_c ET \\ \frac{dE}{dt} &= s_e - d_e E + \alpha_0 k_0 N \frac{T}{T + g} - \gamma_e ET \\ \frac{dN}{dt} &= s_0 - d_0 N - k_0 N \frac{T}{T + g}\end{aligned}\tag{1}$$

This is system of nonlinear first order ODE With 3 initial conditions as given below. With regards to classification, the system has dependent variable t with independent variables as T, E, N and is autonomous with non-linear terms of $ET, NT/(T + g), T^2$. For the solution we use $T(0) = 10000, E(0) = 120, N(0) = 1510$ to alternate the state of Equilibrium. We tested various initial conditions such that we can see the permutation in the solution when the Tumor is not at Equilibrium.

The units of T, E , and N in units of *cells/ μ l* with coefficients above in (1) is given as

$$a = 0.018, \quad b = 1/230000, \quad \gamma_c = 0.005, \quad \gamma_e = 0.0001, \quad g = 720$$

$$k_0 = 0.062, \quad \alpha_0 = 100, \quad s_e = 36, \quad d_e = 0.30, \quad s_0 = 0.37, \quad d_0 = 0.23$$

Assume the power series solution has the form [3]

$$T(t) = \sum_{0 \leq n \leq \infty} f_n t^n, \quad E(t) = \sum_{0 \leq n \leq \infty} l_n t^n, \quad N(t) = \sum_{0 \leq n \leq \infty} m_n t^n, \quad |t| < R\tag{2}$$

So we have the system of nonlinear ordinary differential equation as

$$\begin{aligned}\sum_{0 \leq n \leq \infty} (n+1)f_{n+1}t^n &= a \sum_{0 \leq n \leq \infty} f_n t^n \left(1 - b \sum_{0 \leq n \leq \infty} f_n t^n\right) - \gamma_c \left(\sum_{0 \leq n \leq \infty} l_n t^n\right) \left(\sum_{0 \leq n \leq \infty} f_n t^n\right) \\ \sum_{0 \leq n \leq \infty} (n+1)l_{n+1}t^n &= s_e - d_e \sum_{0 \leq n \leq \infty} l_n t^n + \alpha_0 k_0 \sum_{0 \leq n \leq \infty} m_n t^n \left(\frac{\sum_{0 \leq n \leq \infty} f_n t^n}{\sum_{0 \leq n \leq \infty} f_n t^n + g}\right) - \gamma_e \left(\sum_{0 \leq n \leq \infty} l_n t^n\right) \left(\sum_{0 \leq n \leq \infty} f_n t^n\right) \\ \sum_{0 \leq n \leq \infty} (n+1)m_{n+1}t^n &= s_0 - d_0 \sum_{0 \leq n \leq \infty} m_n t^n - k_0 \sum_{0 \leq n \leq \infty} m_n t^n \left(\frac{\sum_{0 \leq n \leq \infty} f_n t^n}{\sum_{0 \leq n \leq \infty} f_n t^n + g}\right)\end{aligned}\tag{3}$$

Applying the Cauchy product on the dT/dt equation yields

$$\sum_{0 \leq n \leq \infty} (n+1)f_{n+1}t^n = a \sum_{0 \leq n \leq \infty} f_n t^n - ab \sum_{0 \leq n \leq \infty} \left(\sum_{0 \leq j \leq n} f_j f_{n-j} \right) t^n - \gamma_c \sum_{0 \leq n \leq \infty} \left(\sum_{0 \leq j \leq n} l_j f_{n-j} \right) t^n \quad (4)$$

Notice that the $N \frac{T}{T+g}$ term appears two times, so we can let $\tilde{f}_n = g + f_n$ with $\tilde{f}_0 = g + f_0$ and $\sum_{0 \leq n \leq \infty} H_n t^n = N(1 - g/(T+g)) = N - gN/(T+g)$ and treat H_n separately.

$$\begin{aligned} \sum_{0 \leq n \leq \infty} H_n t^n &= \sum_{0 \leq n \leq \infty} m_n t^n - g \left(\frac{\sum_{0 \leq n \leq \infty} m_n t^n}{\sum_{0 \leq n \leq \infty} \tilde{f}_n t^n} \right) \\ &= \sum_{0 \leq n \leq \infty} m_n t^n - g \sum_{0 \leq n \leq \infty} p_n t^n \end{aligned} \quad (5)$$

Where for $n \geq 1$

$$p_n = \frac{1}{\tilde{f}_0} \left(m_n - \sum_{1 \leq j \leq n} \tilde{f}_j p_{n-j} \right), \quad p_0 = \frac{m_0}{\tilde{f}} \approx 0.1408582, \quad p_1 = \frac{m_1 - f_1 p_0}{\tilde{f}} \approx 0.0360667 \quad (6)$$

with $\tilde{f}_0 = g + T(0)$. From H_n in (5) we can write dE/dt and dN/dt as

$$\sum_{0 \leq n \leq \infty} (n+1)l_{n+1}t^n = s_e - d_e \sum_{0 \leq n \leq \infty} l_n t^n + \alpha_0 k_0 \sum_{0 \leq n \leq \infty} H_n t^n - \gamma_e \sum_{0 \leq n \leq \infty} \left(\sum_{0 \leq j \leq n} l_j f_{n-j} \right) t^n \quad (7)$$

$$\sum_{0 \leq n \leq \infty} (n+1)m_{n+1}t^n = s_0 - d_0 \sum_{0 \leq n \leq \infty} m_n t^n - k_0 \sum_{0 \leq n \leq \infty} H_n t^n \quad (8)$$

From the above, we deduce the recurrence relation for $n \geq 1$ and the s_0, s_e constants were accounted for in $n = 0$

$$\begin{aligned} f_{n+1} &= \frac{1}{n+1} \left[a f_n - \sum_{1 \leq j \leq n} f_{n-j} (a b f_j + \gamma_c l_j) \right], \quad f_0 = 10000, \quad f_1 = \frac{dT}{dt}(0) \approx -5827.826 \\ l_{n+1} &= \frac{1}{n+1} \left[-d_e l_n + \alpha_0 k_0 H_n - \gamma_e \sum_{1 \leq j \leq n} l_j f_{n-j} \right], \quad l_0 = 120, \quad l_1 = \frac{dE}{dt}(0) \approx 8613.209 \\ m_{n+1} &= \frac{1}{n+1} [-d_0 m_n - k_0 H_n], \quad m_0 = 1510, \quad m_1 = \frac{dN}{dt}(0) \approx -434.26208955 \\ H_n &= m_n - g p_n, \quad H_0 = m_0 - g p_0 \approx 1408.58, \quad H_1 = m_1 - \frac{g}{f_0 + g} (m_1 - f_1 p_0) \approx -460.2301156 \end{aligned} \quad (9)$$

Notice that there are multiple ways to compute the first two terms in some of the relations, which is used in the code verify our derivations above. The exact values of the initial conditions and $p_1 f_1, l_1, m_1, H_1$ is used in the computations. The computations for exact values is given below.

$$\begin{aligned} f_0 &= T(0), \quad f_1 = a T_0 (1 - b T_0) - \gamma_c E_0 T_0 \\ l_0 &= E(0), \quad l_1 = s_e - d_e E_0 + \alpha_0 k_0 N_0 \frac{T_0}{T_0 + g} - \gamma_e E_0 T_0 \\ m_0 &= N(0), \quad m_1 = s_0 - d_0 N_0 - k_0 N_0 \frac{T_0}{T_0 + g}, \quad \tilde{f}_0 = g + T_0 \\ p_0 &= \frac{m_0}{\tilde{f}_0} = \frac{N_0}{g + T_0}, \quad p_1 = \frac{1}{\tilde{f}_0} (m_1 - f_1 p_0) = \frac{1}{g + T_0} \left(m_1 - f_1 \frac{N_0}{g + T_0} \right) \\ H_0 &= m_0 - g p_0 = \frac{N_0 T_0}{g + T_0}, \quad H_1 = m_1 - g p_1 = m_1 - \frac{g}{g + T_0} \left(m_1 - f_1 \frac{N_0}{g + T_0} \right) \end{aligned} \quad (10)$$

We have the power series solutions and numerical solutions plotted as follows.

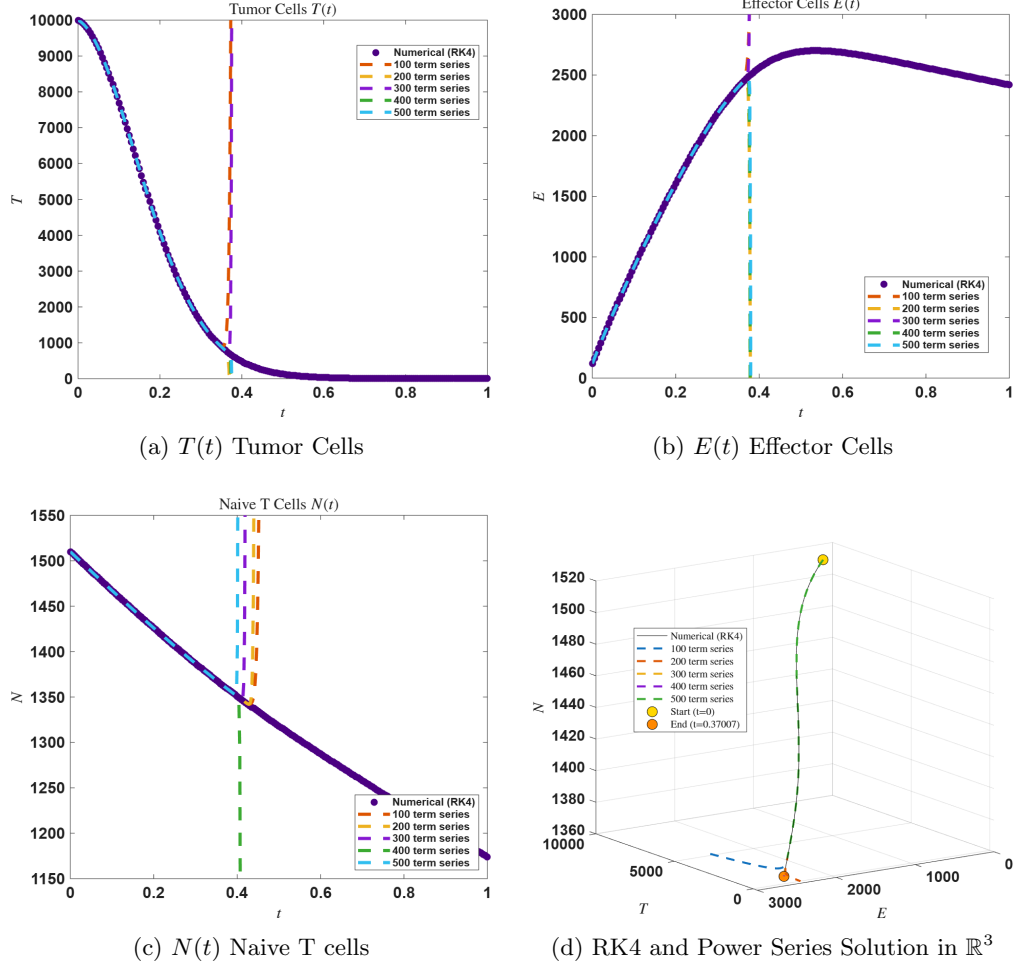


Figure 1: Power series approximations of $T(t)$, $E(t)$, and $N(t)$ compared against the RK4 numerical solution over $t \in [0, 1]$, with the 3D phase portrait in $\mathbb{R}^3$ restricted to $t \in [0, 0.37007]$. Subfigures (a)–(c) show successive truncated power series at every 100 terms up to $N = 500$, converging to the RK4 solution within the radius of convergence $R \approx 0.37$ before diverging beyond it. A rough estimates using the root test is shown below. Subfigure (d) shows the 3D trajectory $T-E-N$ with truncated series at every 100 terms up to $N = 500$, with gold and orange markers denoting the start and end points at $t = 0$ and $t = 0.37007$ respectively.

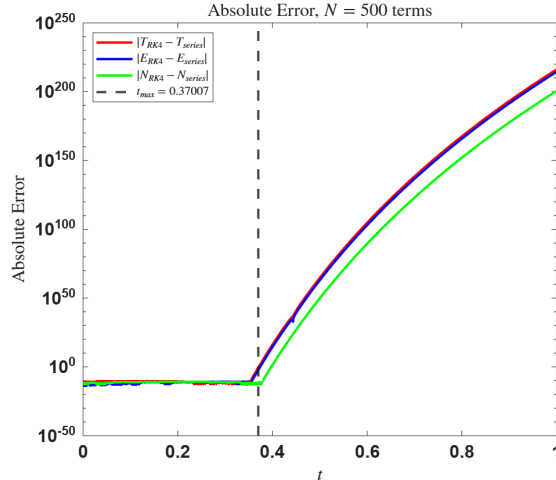


Figure 2: Absolute Error $|f_{RK4} - f_{series}| \leq \epsilon$ for T, E, N within $R \approx t_{max}$

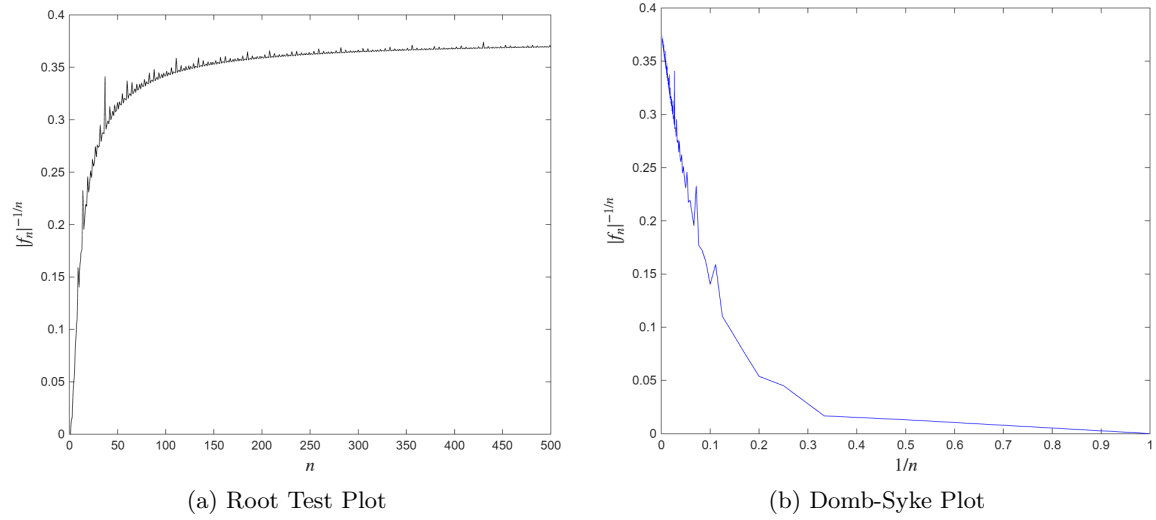


Figure 3: Root test and Domb-Sykes plot of the f_n coefficients, with the y-intercept extrapolating to $1/R$, estimating the radius of convergence $R \approx 0.37$.

*From the two plots above, We were not able to get the **exact** or machine ϵ accurate radius since the behavior was not linear towards the end and this is be done in future improvements with continuations of the power series.*

R Plots

Some R-Studio numerical solutions were tested in the process for intuition.

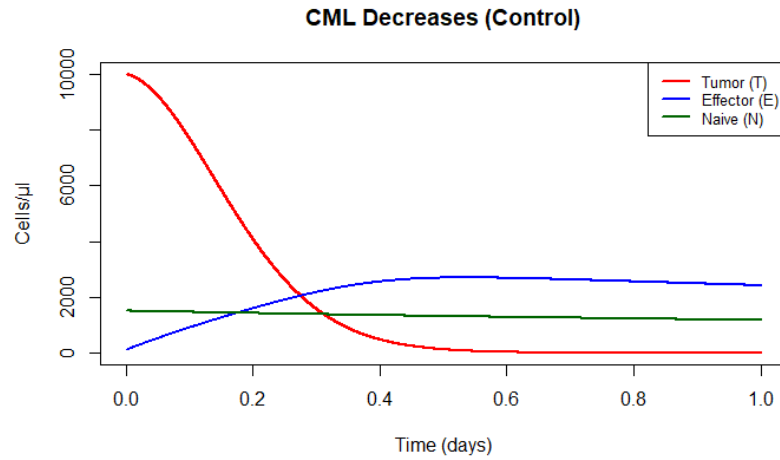


Figure 4: $T(t), E(t), N(t)$ with $t \in [0, 1]$

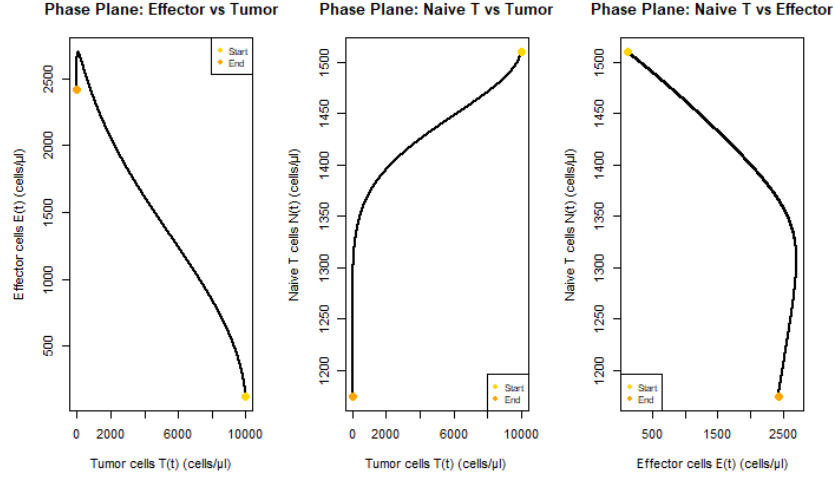


Figure 5: **Left**: $E(t)$ vs $T(t)$: As the tumor grows, effector cells are rapidly depleted by the mass action killing term $\gamma_e ET$, showing the immune system is unable to sustain a response over $t \in [0, 0.299]$. **Middle**: $N(t)$ vs $T(t)$: The near-linear depletion of naive T cells with tumor growth reflects the recruitment term $k_0 N \frac{T}{T+g}$ continuously drawing naive cells out of circulation, weakening the upstream supply of new effectors. **Right**: $N(t)$ vs $E(t)$: Both immune populations decline together but at different rates effectors drop faster due to the additional loss term $\gamma_e ET$, while naive cells deplete more gradually through recruitment and natural death $d_0 N$ alone.

Analytic Continuations

To observe if the solution reaches some steady-state behavior, we run the numerical solution for a larger t such as $t = 15$.

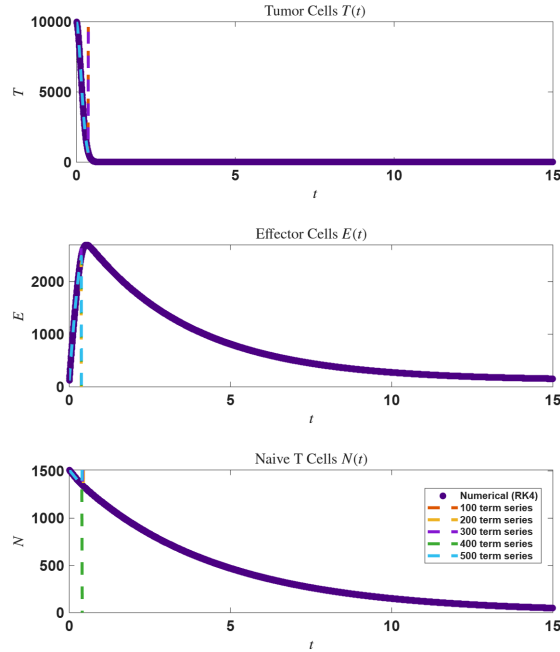


Figure 6: T, E, N plot for $t \in [0, 15]$

From the figure above, it is clear enough T approaches 0 for $t \geq 15$, and E, N is decreasing towards 0 at a similar rate. For long term behavior of the solution as $t \rightarrow \infty$ is equivalent to setting the derivatives to 0 and solving for $T_\infty, E_\infty, N_\infty$ yields

$$\begin{aligned}
T_\infty &= 0 \\
E_\infty &= \frac{a - abT_\infty}{\gamma_c} = \frac{a}{\gamma_c} \\
N_\infty &= \frac{s_0(g + T_\infty)}{d_0(g + T_\infty) + k_0T_\infty} = \frac{s_0}{d_0}
\end{aligned} \tag{11}$$

From (11) we know the asymptotic behaviors $T \sim 0, E \sim a/\gamma_c, N \sim s_0/d_0$. We suspect the following ordinate re-summation for T with $T_\infty = 0$ assuming the solution (2).

$$\begin{aligned}
T(t) = T_\infty + \frac{1}{\sum_{0 \leq n \leq \infty} Q_n t^n} &= \sum_{0 \leq n \leq \infty} f_n t^n \Rightarrow \frac{1}{T(t)} = \frac{-1}{f_0} \sum_{0 \leq n \leq \infty} \left(\sum_{1 \leq j \leq n} f_j Q_{n-j} \right) t^n, \quad Q_0 = \frac{1}{f_0} \\
T(t) &= \left(\sum_{0 \leq n \leq \infty} Q_n t^n \right)^{-1} = \left\{ \frac{-1}{f_0} \sum_{0 \leq n \leq \infty} \left(\sum_{1 \leq j \leq n} f_j Q_{n-j} \right) t^n \right\}^{-1}, \quad Q_0 = \frac{1}{f_0}
\end{aligned} \tag{12}$$

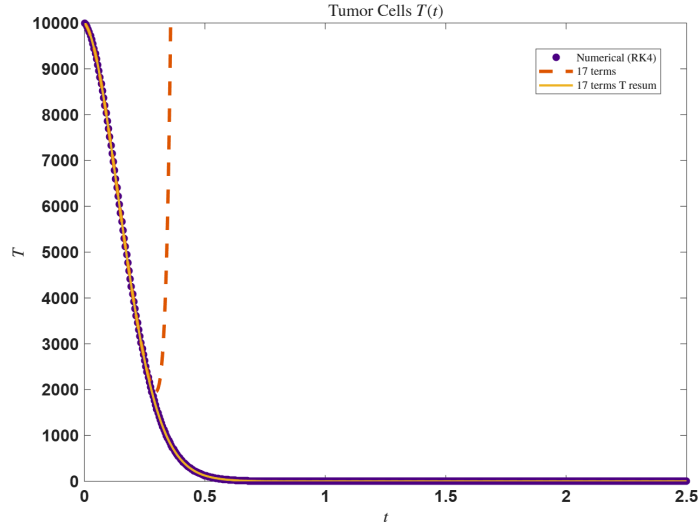


Figure 7: Ordinate Re-summation for $T(t)$ Truncated at $N = 17$

We continue the re-summation process for $E(t)$ using ordinate re-summation or a $[0/N_\infty]$ Padé approximant with asymptotics behavior $E_\infty = a/\gamma_c$.

$$E(t) = \sum_{0 \leq n \leq \infty} l_n t^n = a/\gamma_c + \frac{1}{\sum_{0 \leq n \leq \infty} L_n t^n} \tag{13}$$

For notational convention let $\sum_{0 \leq n \leq \infty} \tilde{l}_n = \underbrace{(l_0 - a/\gamma_c)}_{\tilde{l}_0} + \sum_{1 \leq n \leq \infty} l_n$, then we can obtain a re-summation for $E(t)$

$$E(t) = \left(\sum_{0 \leq n \leq \infty} L_n t^n \right)^{-1} = \left\{ -\frac{1}{\tilde{l}_0} \sum_{0 \leq n \leq \infty} \left(\sum_{1 \leq j \leq n} \tilde{l}_j L_{n-j} \right) t^n \right\}^{-1}, \quad L_0 = -\tilde{l}_0^{-1} \tag{14}$$

The results for (14) is not sufficient, thus we then attempt another re-summation via a Padé asymptotics motivation. The Padé approximant of order $[N/M]$ for $E(t)$ is constructed from the truncated power series

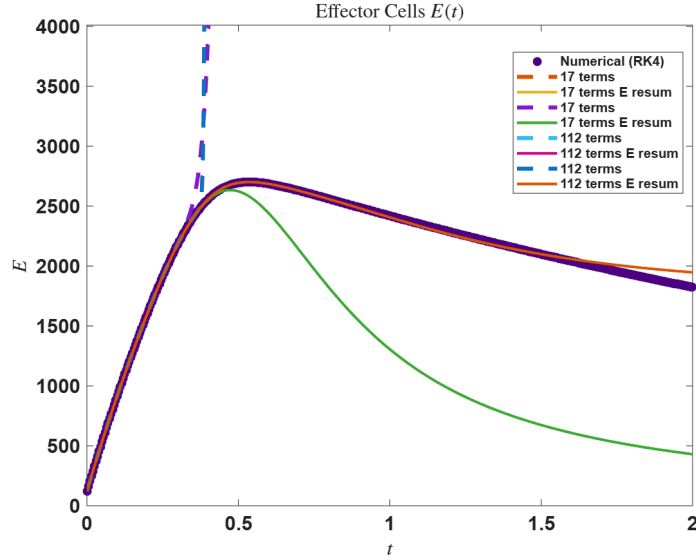


Figure 8: Re-summation of $E_{Pade}(t)$

$$E(t) = \sum_{n=0}^{N+M} l_n t^n + \mathcal{O}(t^{N+M+1}) \quad (15)$$

by expressing it as a rational function

$$E_{[N/M]}(t) = \frac{P_N(t)}{Q_M(t)} = \frac{\sum_{k=0}^N A_k t^k}{1 + \sum_{k=1}^M B_k t^k} \quad (16)$$

where the coefficients $\{A_k\}_{k=1}^N$ and $\{B_k\}_{k=1}^M$ are determined by the requirement that the Taylor expansion of P_N/Q_M matches the original series up to order $N+M$:

$$\left(\sum_{k=0}^N A_k t^k \right) - \left(\sum_{n=0}^{N+M} l_n t^n \right) \left(1 + \sum_{k=1}^M B_k t^k \right) = \mathcal{O}(t^{N+M+1}) \quad (17)$$

with $A_0 = l_0 = E(0)$. Equating coefficients for t^{N+1} through t^{N+M} yields a linear system for the denominator coefficients:

$$\begin{bmatrix} l_N & l_{N-1} & \cdots & l_{N-M+1} \\ l_{N+1} & l_N & \cdots & l_{N-M+2} \\ \vdots & \vdots & \ddots & \vdots \\ l_{N+M-1} & l_{N+M-2} & \cdots & l_N \end{bmatrix} \begin{bmatrix} B_1 \\ B_2 \\ \vdots \\ B_M \end{bmatrix} = - \begin{bmatrix} l_{N+1} \\ l_{N+2} \\ \vdots \\ l_{N+M} \end{bmatrix} \quad (18)$$

Once the B_k are known, the numerator coefficients follow from the convolution

$$A_n = l_n + \sum_{j=1}^{\min(n,M)} B_j l_{n-j}, \quad n = 1, \dots, N \quad (19)$$

has the property that as $t \rightarrow \infty$,

$$E_{[N/M]}(t) \sim \frac{A_N}{B_M} t^{N-M} \quad (20)$$

Next, The attempt for re-summation for $N(t)$ involved ordinate re-summation, shifted Dirichlet, Pade Approximants, and Gauge Resummation. The best results we achieved was through the Shifted Dirichlet.

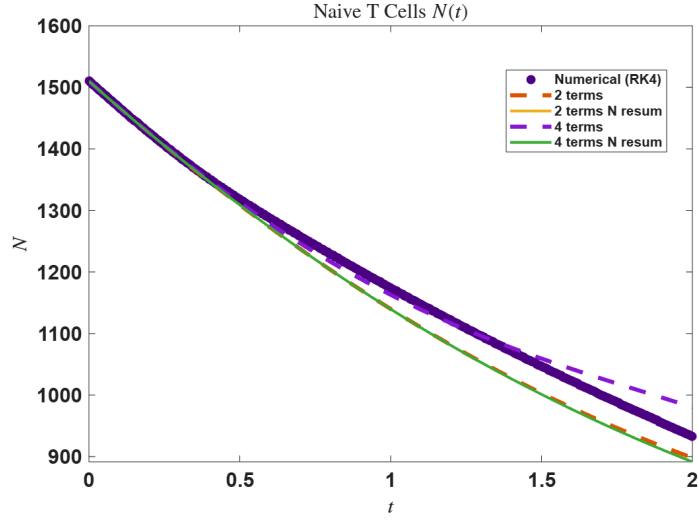


Figure 9: Shifted Dirichlet of $N_{Dir}(t)$

$$N_{dir}(t) = m_0 + \sum_{1 \leq n \leq N} A_n (e^{d_0 t} - 1)^n, \quad A_n = \sum_{1 \leq j \leq n} \frac{j! m_j (-1)^{n-j} S(n, j)}{-d_0^j n!} \quad (21)$$

here m_j are the coefficients in (9) and $S(n, j)$ are the Stirling numbers of the second kind.

Coefficient Signs

From $\text{SIGN}(\cdot)$ on f_n, l_n, m_n we see that for n up to 700 there is no apparent sign pattern for all three coefficients. Hence all corollaries of Pringsheim's theorem can not be applied to find the closest singularity.

```
>> sign(f)
ans =
-1    -1     1     1    -1    -1     1     1    -1     1     1    -1     1     1    -1
-1     1    -1    -1     1     1

>> sign(l)
ans =
1     -1     1     1    -1    -1     1     1    -1     1     1    -1     1     1    -1
-1     1    -1    -1     1     1

>> sign(m)
ans =
-1     1     1    -1     1    -1     1    -1     1    -1    -1     1    -1     1    -1
1     1    -1     1    -1     1
```


Computations and Graphing

R Studio - Chuck 1

```
```{r}

#install.packages("deSolve")
#library(deSolve)

tumor_model <- function(t, state, parameters) {
 with(as.list(c(state, parameters)), {

 dT <- a*T*(1 - b*T) - gamma_c*E*T

 dE <- s_e - d_e*E + alpha_n*k_n*N*T/(T + g) - gamma_e*E*T

 dN <- s_n - d_n*N - k_n*N*T/(T + g)

 list(c(dT, dE, dN))
 })
}

params_fig6 <- c(
 a = 0.018,
 b = 1/230000,
 gamma_c = 0.005,
 gamma_e = 0.0001,
 g = 720,
 k_n = 0.062,
 alpha_n = 100,
 s_e = 36,
 d_e = 0.30,
 s_n = 0.37,
 d_n = 0.23
)

state_fig6 <- c(T = 10000, E = 120, N = 1510)

times <- seq(0, 1, by = 0.0001)
out <- ode(
 y = state_fig6,
 times = times,
 func = tumor_model,
 parms = params_fig6,
 method = "lsoda"
)

out <- as.data.frame(out)

out_fig6 <- as.data.frame(ode(y = state_fig6, times = times,
 func = tumor_model, parms = params_fig6,
 method = "lsoda"))
```
```

```

#plotting
```{R}
plot(out_fig6$time, out_fig6$T,
 type = "l", col = "red", lwd = 2,
 xlab = "Time (days)", ylab = "Cells/ μ l",
 main = "CML Decreases (Control)",
 ylim = range(out_fig6[, c("T","E","N")]))
lines(out_fig6$time, out_fig6$E, col = "blue", lwd = 2)
lines(out_fig6$time, out_fig6$N, col = "darkgreen", lwd = 2)
legend("topright",
 legend = c("Tumor (T)", "Effector (E)", "Naive (N)"),
 col = c("red", "blue", "darkgreen"),
 lty = 1, lwd = 2, cex = 0.8)

Phase Plane: Effector E vs Tumor T
plot(out_fig6T, out_fig6E,
 type = "l", lwd = 2,
 xlab = "Tumor cells T(t) (cells/ μ l)",
 ylab = "Effector cells E(t) (cells/ μ l)",
 main = "Phase Plane: Effector vs Tumor")
points(out_fig6$T[1], out_fig6$E[1],
 pch = 19, col = "gold", cex = 1.5)
points(out_fig6$T[nrow(out_fig6)], out_fig6$E[nrow(out_fig6)],
 pch = 19, col = "orange", cex = 1.5)
legend("topright", legend = c("Start", "End"),
 pch = 19, col = c("gold", "orange"), cex = 0.8)

Phase Plane: Naive N vs Tumor T
plot(out_fig6T, out_fig6N,
 type = "l", lwd = 2,
 xlab = "Tumor cells T(t) (cells/ μ l)",
 ylab = "Naive T cells N(t) (cells/ μ l)",
 main = "Phase Plane: Naive T vs Tumor")
points(out_fig6$T[1], out_fig6$N[1],
 pch = 19, col = "gold", cex = 1.5)
points(out_fig6$T[nrow(out_fig6)], out_fig6$N[nrow(out_fig6)],
 pch = 19, col = "orange", cex = 1.5)
legend("bottomright", legend = c("Start", "End"),
 pch = 19, col = c("gold", "orange"), cex = 0.8)

Phase Plane: Naive N vs Effector E
plot(out_fig6E, out_fig6N,
 type = "l", lwd = 2,
 xlab = "Effector cells E(t) (cells/ μ l)",
 ylab = "Naive T cells N(t) (cells/ μ l)",
 main = "Phase Plane: Naive T vs Effector")
points(out_fig6$E[1], out_fig6$N[1],
 pch = 19, col = "gold", cex = 1.5)
points(out_fig6$E[nrow(out_fig6)], out_fig6$N[nrow(out_fig6)],
 pch = 19, col = "orange", cex = 1.5)
legend("bottomleft", legend = c("Start", "End"),
 pch = 19, col = c("gold", "orange"), cex = 0.8)
```

```

```

#Computing Initial Condition
'''{r}
a      = 0.018
b      = 1/230000
gamma_c = 0.005
gamma_e = 0.0001
g      = 720
k_n    = 0.062
alpha_n = 100
s_e    = 36
d_e    = 0.30
s_n    = 0.37
d_n    = 0.23
f_0 <- 10000
l_0 <- 120
m_0 <- 1510
options(digits = 20)

(H_0 <- m_0 * f_0 / (f_0 + g))
(f_1 <- a*f_0*(1 - b*f_0) - gamma_c * f_0 * l_0 )
(l_1H <- s_e - d_e* l_0 + alpha_n* k_n*m_0*(f_0/(f_0 + g))- gamma_e*f_0*l_0)
(l_1 <- s_e - d_e*l_0 + alpha_n*k_n*H_0 - gamma_e*l_0*f_0)
(m_1 <- s_n - d_n*m_0 - k_n*H_0)
(m_1a <- s_n - d_n*m_0 - k_n *m_0 *(f_0/(f_0 + g)))
cat("p_0 : " , (p_0 <- m_0 / ( f_0 + g)))
(p_1 <-(m_1 - f_1 * p_0 )/ (f_0 + g) )
(H_0 <- m_0 - g* p_0 )
(H_1 <- m_1 - g* (m_1 - f_1 *p_0)/(f_0 + g))
'''

```

```

‘‘{r}
cat("H_0 =", (H_0 <- m_0 * f_0 / (f_0 + g)), "\n")

cat("f_1 =", (f_1 <- a*f_0*(1 - b*f_0) - gamma_c * f_0 * l_0), "\n")

cat("l_1 (via H_0 directly) =", (l_1H <- s_e - d_e*l_0 + alpha_n*k_n*m_0*(f_0/(f_0 + g))
- gamma_e*f_0*l_0), "\n")

cat("l_1 (via H_0 stored)   =", (l_1  <- s_e - d_e*l_0 + alpha_n*k_n*H_0
- gamma_e*l_0*f_0), "\n")

cat("m_1 (via H_0 stored)   =", (m_1  <- s_n - d_n*m_0 - k_n*H_0), "\n")

cat("m_1 (via H_0 directly) =", (m_1a <- s_n - d_n*m_0 - k_n*m_0*(f_0/(f_0 + g))), "\n")

cat("p_0 =", (p_0 <- m_0 / (f_0 + g)), "\n")

cat("p_1 =", (p_1 <- (m_1 - f_1*p_0) / (f_0 + g)), "\n")

cat("H_0 (via p_0) =", (H_0_alt <- m_0 - g*p_0), "\n")

cat("H_1 =", (H_1 <- m_1 - g*(m_1 - f_1*p_0)/(f_0 + g)), "\n")
’’

```

References

- [1] Helen Moore and Natasha K Li. A mathematical model for chronic myelogenous leukemia (cml) and t cell interaction. *Journal of theoretical biology*, 227(4):513–523, 2004.
- [2] Anne Talkington, Claudia Dantoin, and Rick Durrett. Ordinary differential equation models for adoptive immunotherapy. *Bulletin of mathematical biology*, 80(5):1059–1083, 2018.
- [3] Nathaniel S. Barlow and Steven J. Weinstein. *Power Series Solutions to Nonlinear Ordinary Differential Equations and Related Problems of Physics, Engineering, and Life Sciences*, volume 208 of *Other Titles in Applied Mathematics*. Society for Industrial and Applied Mathematics, Philadelphia, PA, 2025.